

Week 9 — Verification Output (Auditing Requirement)

Goal

Add verification after solving and show it in every run.

Required Outputs

- ☐ Verification implemented (numeric residual/error OR symbolic substitution/back-check)
- ☐ Verification displayed in UI trail for every computation
- ☐ At least 5 verification tests documented

Evidence to Attach

- ☐ Screenshot: trail includes a VERIFICATION section with computed check result

EVIDENCE TO ATTACH WITH TEST CASES PICTURES

Test	Method	Function	Initial Guess	Root Found	Residual	Status
1	Newton-Raphson	$x^2 - 2$	1.5	1.4142135623 7	~0	✓ VERIFIED
2	Newton-Raphson	$\cos(x) - x$	1.0	0.7390851332 1	~0	✓ VERIFIED
3	Secant	$x^3 - 3x + 1$	1.5, 2.0	1.5320888862 5	~0	✓ VERIFIED
4	Newton-Raphson	$x^3 - 3x + 1$	2.0	1.5320888862 4	~0	✓ VERIFIED
5	Secant	$x^2 - 2$	1.0, 3.0	1.4142135623 7	~0	✓ VERIFIED

TEST #1

Newton-Raphson Root Finder v2.0 (Custom/Twitter)

Solver Configuration

Method: Newton-Raphson

Function: $x^2 - 2$

Initial Guess: 1.5

Tolerance: $1e-6$

Max Iterations: 50

[Compute](#) [Clear](#) [Export](#)

Final Root: 1.41421356237

Solution Trail

GIVEN:
Method: Newton-Raphson
Function: $x^2 - 2$
Initial guess: 1.5
Tolerance: $1e-06$
Max iterations: 50

VALIDATION:
1. All inputs valid.

EDGE CASE ANALYSIS:
✓ No edge cases detected. Configuration looks safe.

METHOD:
Newton-Raphson

STOPPING RULES:
1. Stop if $|x_{new} - x| < \text{tolerance}$ ($1e-06$)
2. Stop if iterations exceed max (50)
3. Stop if derivative = 0

STEPS:
1. $i=1, x=1.5000000000, f(x)=2.500e-01, x_{next}=1.4166666667, \text{err}=3.333e-02$
2. $i=2, x=1.4166666667, f(x)=6.944e-03, x_{next}=1.4142116063, \text{err}=2.431e-03$
3. $i=3, x=1.4142116063, f(x)=6.097e-06, x_{next}=1.4142135624, \text{err}=2.124e-06$
4. $i=4, x=1.4142135624, f(x)=4.511e-12, x_{next}=1.4142135624, \text{err}=1.595e-12$

FINAL ANSWER:
Method Used: Newton-Raphson
 $x = 1.41421356237$

STOPPING CONDITION:
SUCCESS: Converged (error $1.595e-12 < 1e-06$)

SUMMARY:
Method: Newton-Raphson
Iterations: 4
Time: 0.0014s
Timestamp: 2026-04-24 18:32:30

VERIFICATION:
Root found: $x = 1.41421356237$
Residual: $f(x) = 4.440892e-16$
Absolute residual: $|f(x)| = 4.440892e-16$
✓ VERIFIED: $|f(x)| = 4.440892e-16 < \text{tolerance}$ ($1e-06$)
The root satisfies $f(x) = 0$ within the specified tolerance.

Theme: [Dark](#) | Trail shows: steps (black) | success (green) | errors/warnings (red) [About](#)

TEST#2

Newton-Raphson Root Finder v2.0 (Custom/Twitter)

Solver Configuration

Method: Newton-Raphson

Function: $\cos(x)$

Initial Guess: 1

Tolerance: $1e-6$

Max Iterations: 50

[Compute](#) [Clear](#) [Export](#)

Final Root: 0.739085133215

Solution Trail

GIVEN:
Method: Newton-Raphson
Function: $\cos(x)$
Initial guess: 1
Tolerance: $1e-06$
Max iterations: 50

VALIDATION:
1. All inputs valid.

EDGE CASE ANALYSIS:
✓ No edge cases detected. Configuration looks safe.

METHOD:
Newton-Raphson

STOPPING RULES:
1. Stop if $|x_{new} - x| < \text{tolerance}$ ($1e-06$)
2. Stop if iterations exceed max (50)
3. Stop if derivative = 0

STEPS:
1. $i=1, x=1.0000000000, f(x)=-4.597e-01, x_{next}=0.7803638678, \text{err}=3.496e-01$
2. $i=2, x=0.7803638678, f(x)=-1.892e-02, x_{next}=0.7391128909, \text{err}=1.125e-02$
3. $i=3, x=0.7391128909, f(x)=-4.646e-05, x_{next}=0.7390851334, \text{err}=2.776e-05$
4. $i=4, x=0.7390851334, f(x)=-2.847e-10, x_{next}=0.7390851332, \text{err}=1.701e-10$

FINAL ANSWER:
Method Used: Newton-Raphson
 $x = 0.739085133215$

STOPPING CONDITION:
SUCCESS: Converged (error $1.701e-10 < 1e-06$)

SUMMARY:
Method: Newton-Raphson
Iterations: 4
Time: 0.0014s
Timestamp: 2026-04-24 18:34:00

VERIFICATION:
Root found: $x = 0.739085133215$
Residual: $f(x) = 0.000000e+00$
Absolute residual: $|f(x)| = 0.000000e+00$
✓ VERIFIED: $|f(x)| = 0.000000e+00 < \text{tolerance}$ ($1e-06$)
The root satisfies $f(x) = 0$ within the specified tolerance.

Theme: [Dark](#) | Trail shows: steps (black) | success (green) | errors/warnings (red) [About](#)

TEST#3

Newton-Raphson Root Finder v2.0 (Custom/Tester)

Newton-Raphson Root Finder

Solver Configuration

Method:
Secant

Function:
 $x^3 - 3x + 1$

Initial Guess:
1.5

Tolerance:
 $1e-6$

Max Iterations:
50

Compute

Clear

Export

Second Guess:
2.0

Final Root:
1.53208888621

Solution Trail

GIVEN:
Method: Secant
Function: $x^3 - 3x + 1$
Initial guess: 1.5
Second guess: 2.0
Tolerance: $1e-6$
Max Iterations: 50

VALIDATION:
1. All inputs valid.

EDGE CASE ANALYSIS:
✓ No edge cases detected. Configuration looks safe.

METHOD:
Secant

STOPPING RULES:
1. Stop if $|x_{new} - x| < \text{tolerance}$ ($1e-6$)
2. Stop if iterations exceed max (50)
3. Stop if denominator = 0

STEPS:
2. $i=1, x_{prev}=1.5000000000, x=2.0000000000, x_{next}=1.5200000000, err=4.800e-01$
3. $i=2, x_{prev}=2.0000000000, x=1.5200000000, x_{next}=1.5278888133, err=7.589e-03$
4. $i=3, x_{prev}=1.5200000000, x=1.5278888133, x_{next}=1.5321517070, err=4.543e-03$
5. $i=4, x_{prev}=1.5278888133, x=1.5321517070, x_{next}=1.5320888634, err=6.314e-05$
6. $i=5, x_{prev}=1.5321517070, x=1.5320888634, x_{next}=1.5320888862, err=3.228e-07$

FINAL ANSWER:
Method Used: Secant
 $x = 1.53208888621$

STOPPING CONDITION:
SUCCESS: Converged (error $3.228e-07 < 1e-6$)

SUMMARY:
Method: Secant
Iterations: 5
Time: 0.0021s
Timestamp: 2024-04-24 18:37:33

VERIFICATION:
Root found: $x = 1.53208888621$
Residual: $f(x) = -9.319456e-11$
Absolute residual: $|f(x)| = 9.319456e-11$
✓ VERIFIED: $|f(x)| = 9.319456e-11 < \text{tolerance}$ ($1e-6$)
The root satisfies $f(x) = 0$ within the specified tolerance.

Theme: Dark

Trail shows: steps (black) | success (green) | errors/warnings (red)

About

TEST#4

Solver Configuration

Method:

Newton-Raphson

Function:

$x^3 - 3x + 1$

Initial Guess:

2.0

Tolerance:

$1e-6$

Max Iterations:

50

Compute

Clear

Export

Final Root:

1.53208888624

Solution Trail

GIVEN:

Method: Newton-Raphson
Function: $x^3 - 3x + 1$
Initial guess: 2.0
Tolerance: $1e-06$
Max iterations: 50

VALIDATION:

1. All inputs valid.

EDGE CASE ANALYSIS:

✓ No edge cases detected. Configuration looks safe.

METHOD:

Newton-Raphson

STOPPING RULES:

1. Stop if $|x_{new} - x| < \text{tolerance}$ ($1e-06$)
2. Stop if iterations exceed max (50)
3. Stop if derivative = 0

STEPS:

2. $i=1, x=1.0000000000, f(x)=3.000e+00, x_{next}=1.6666666667, err=3.333e-01$
3. $i=2, x=1.6666666667, f(x)=6.286e-03, x_{next}=1.9496111111, err=1.131e-01$
4. $i=3, x=1.9496111111, f(x)=6.504e-02, x_{next}=1.5323901619, err=1.622e-02$
5. $i=4, x=1.5323901619, f(x)=1.218e-03, x_{next}=1.5320888864, err=3.012e-04$
6. $i=5, x=1.5320888864, f(x)=4.170e-07, x_{next}=1.5320888862, err=1.032e-07$

FINAL ANSWER:

Method Used: Newton-Raphson
 $x = 1.53208888624$

STOPPING CONDITION:

SUCCESS: Converged (error $1.032e-07 < 1e-06$)

SUMMARY:

Method: Newton-Raphson
Iterations: 5
Time: 0.0023s
Timestamp: 2026-04-24 18:40:46

VERIFICATION:

Root found: $x = 1.53208888624$
Residual: $f(x) = 4.840572e-14$
Absolute residual: $|f(x)| = 4.840572e-14$
✓ VERIFIED: $|f(x)| = 4.840572e-14 < \text{tolerance}$ ($1e-06$)
The root satisfies $f(x) = 0$ within the specified tolerance.

TEST#5

Solver Configuration

Method:

Secant

Function:

$x^2 - 2$

Initial Guess:

1.0

Tolerance:

$1e-6$

Max Iterations:

50

Compute

Clear

Export

Second Guess:

3.0

Final Root:

1.41421356238

Solution Trail

GIVEN:

Method: Secant
Function: $x^2 - 2$
Initial guess: 1.0
Second guess: 3.0
Tolerance: $1e-06$
Max iterations: 50

VALIDATION:

1. All inputs valid.

EDGE CASE ANALYSIS:

✓ No edge cases detected. Configuration looks safe.

METHOD:

Secant

STOPPING RULES:

1. Stop if $|x_{new} - x| < \text{tolerance}$ ($1e-06$)
2. Stop if iterations exceed max (50)
3. Stop if denominator = 0

STEPS:

2. $i=1, x_{prev}=1.0000000000, x=3.0000000000, x_{next}=1.2500000000, err=1.750e+00$
3. $i=2, x_{prev}=3.0000000000, x=1.2500000000, x_{next}=1.3529411765, err=1.029e-01$
4. $i=3, x_{prev}=1.2500000000, x=1.3529411765, x_{next}=1.4180790960, err=6.514e-02$
5. $i=4, x_{prev}=1.3529411765, x=1.4180790960, x_{next}=1.414280883, err=9.531e-03$
6. $i=5, x_{prev}=1.4180790960, x=1.414280883, x_{next}=1.4142134457, err=0.536e-05$
7. $i=6, x_{prev}=1.414280883, x=1.4142134457, x_{next}=1.4142135624, err=1.167e-07$

FINAL ANSWER:

Method Used: Secant
 $x = 1.41421356238$

STOPPING CONDITION:

SUCCESS: Converged (error $1.167e-07 < 1e-06$)

SUMMARY:

Method: Secant
Iterations: 6
Time: 0.0023s
Timestamp: 2026-04-24 18:41:36

VERIFICATION:

Root found: $x = 1.41421356238$
Residual: $f(x) = 9.971579e-12$
Absolute residual: $|f(x)| = 9.971579e-12$
✓ VERIFIED: $|f(x)| = 9.971579e-12 < \text{tolerance}$ ($1e-06$)
The root satisfies $f(x) = 0$ within the specified tolerance.

Reflection

How did verification improve confidence in your results?

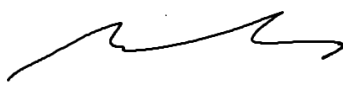
Verification improved our confidence in the results because we didn't just rely on the method stopping—we actually checked if the answer is correct. After getting the root using Newton-Raphson or Secant, we substituted it back into the original function and computed the residual $f(x)f(x)f(x)$. Then we checked if the absolute value $|f(x)||f(x)||f(x)|$ is close to zero and within the given tolerance.

This helped us confirm that the solution really satisfies the equation, not just that the iterations stopped. It also made our system more reliable because we can classify the result as verified, acceptable, or inaccurate depending on how small the residual is.

Another important part is that it helps catch errors. Sometimes, the method can stop due to reaching the maximum iterations or small changes between values, but the answer is still not accurate. With verification, we can detect those cases and warn the user.

Overall, verification gave us an extra layer of checking, so instead of just trusting the computed root, we have actual numerical evidence that the result is correct.

Weekly Sign-off


Group Leader Name/Signature: CHARLES RUSSEL O. SAN JUAN Date: 4/24/2026


Member Name/Signature: CHRISTINA GOMBA Date: 4/24/2026


Member Name/Signature: JOHN MICHAEL TULALIAN Date: 3/24/2026